Determining pore-throat size in Permo-Triassic sandstones from low-frequency electrical spectroscopy

Julian B. T. Scott and Ron D. Barker

School of Geography, Earth and Environmental Sciences, University of Birmingham, Edgbaston, Birmingham, UK

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[1] Hydraulic conductivity or permeability, a knowledge of which is vital in modelling the movement of hydrocarbons in reservoir rocks and contaminants in aquifers, is strongly related to the intergranular pore-throat size in sandstones. For many years attempts have been made to determine permeability from the electrical properties of rocks, but with little success. Here we have measured the complex electrical conductivity of Permo-Triassic sandstone samples over the frequency range 0.0001–1000 Hz and show how parameters determined from the characteristic shape of the spectra correlate closely to pore-throat size determined from mercury injection measurements. *INDEX TERMS:* 5109 Physical Properties of Rocks: Magnetic and electrical properties; 1829 Hydrology: Groundwater hydrology; 0925 Exploration Geophysics: Magnetic and electrical methods; 3675 Mineralogy and Petrology: Sedimentary petrology. **Citation:** Scott, J. B. T., and R. D. Barker, Determining pore-throat size in Permo-Triassic sandstones from low-frequency electrical spectroscopy, *Geophys. Res. Lett.*, *30*(9), 1450, doi:10.1029/2003GL016951, 2003.

1. Introduction

[2] Determining permeability of porous rocks is important in understanding the migration of economic hydrocarbons in reservoir studies, in modelling groundwater flow in hydrogeological studies, and in determining the risk of contamination in pollution studies. Here we show how measurements of electrical resistivity made over a broad frequency range may be correlated directly with the sizes of the throats or pathways between the pore spaces in the rock. As the diameter of pore-throats is intimately related to the permeability [*Bloomfield et al.*, 2001], a simple technique for the field determination of permeability may be possible.

[3] Low-frequency electrical measurements of complex conductivity have been used to detect contaminants in rocks and soils [*Olhoeft*, 1985; *Börner et al.*, 1993; *Vanhala*, 1997] but the causes of the measured electrical phenomena are not fully understood. In non-metallically mineralised sandstone a quadrature or out-of-phase conductivity has its origins in an electrical double layer (EDL) which is present around mineral grains. This EDL consists of a fixed layer of cations, the Stern layer, surrounded by a diffuse mobile layer, the Gouy layer. The passage of an electrical current causes the ions in and around these layers to move relative to each other [*Pengra et al.*, 1999; *Revil*, 1999], resulting in charge build up and a quadrature conductivity. The effect in clean quartz sandstones is small because the EDL is weak. However, pore-lining clays and oxyhydroxides that frequently occur in sandstone often have a negative charge and a large surface area. This increases the quantity of cations in the EDL needed to balance the negative charge, significantly increasing the quadrature conductivity. The time taken for an ion in or around the EDL to return to its equilibrium position after cessation of current flow is known as a relaxation time. The superposition of these relaxation times determines the frequency dependence of the quadrature conductivity when an alternating current is applied [*Vinegar and Waxman*, 1984; *Lesmes and Morgan*, 2001]. Early work on artificial sandstones made from mixtures of clay and glass beads [*Klein and Sill*, 1982] suggested that these relaxation times were dependent on grain or pore size. However, as previous measurements have been made on clean sandstones, or over insufficient frequency ranges, the frequency dependence of the quadrature conductivity was normally ignored. This has led to the application of a constant phase angle model in the low-frequency range, and this contains no information on length scales in the sandstone [*Börner et al.*, 1993; *Pengra et al.*, 1999; *Vinegar and Waxman*, 1984]. Recently, models have been proposed, to account for the observed frequency dependence of the quadrature conductivity based on grain size distributions [*Lesmes and Morgan*, 2001; *Nettelblad and Niklasson*, 1997; *Garrouch*, 1999]. However, these models either only work over limited frequency ranges or do not fit the observed low-frequency electrical phenomena.

[4] Laboratory measurements of the low-frequency complex conductivity spectra including the magnitude and phase angle over the frequency range 0.0001–1000 Hz have been made on samples of Permo-Triassic sandstone from the U.K. This sandstone is an important aquifer rock providing 25% of groundwater abstractions in England and Wales [*Allen et al.*, 1997] and is linked to oil and gas field sandstone basins in the Irish Sea, the North Sea and the English Channel. Standard petrological measurements have been used in order to explain the results of the electrical measurements.

2. Laboratory Methods

[5] Complex conductivity, pore-surface area from nitrogen adsorption, pore-throat size from mercury injection and cation exchange capacity were measured on Permo-Triassic sandstone from boreholes in all of the main Permo-Triassic sandstone areas of the U.K. (Figure 1). Low-frequency, complex conductivity measurements were made on 18 cylindrical core samples, each measuring 35 mm in diameter and 30–70 mm in length, selected in order to obtain a wide distribution of grain size. The samples were then

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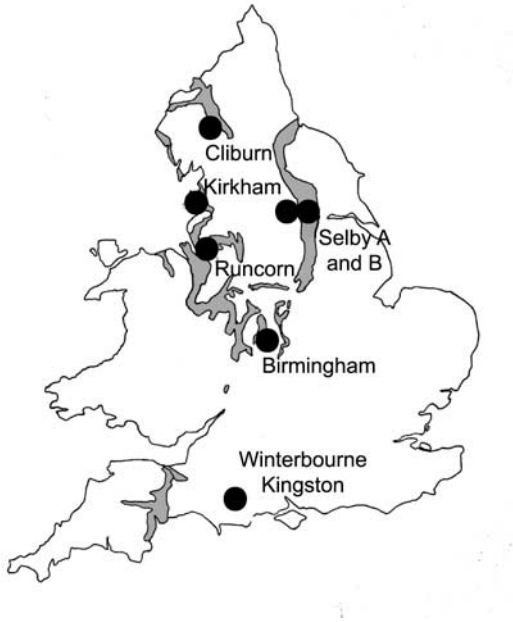


Figure 1. The main Permo-Triassic sandstone outcrops of England and Wales. Sandstone outcrop areas are shaded. Black spots show the boreholes from which sandstone cores used in this study were obtained.

vacuum saturated with a synthetic groundwater, containing 60 mgl^{-1} Ca^{2+}, 30 mgl^{-1} Mg^{2+}, 34 mgl^{-1} Na^{+}, 142 mgl^{-1} Cl^{-}, 120 mgl^{-1} SO_4^{2-} and 30 mgl^{-1} HCO_3^{-}, which closely matches the cation concentrations in the Triassic sandstone aquifer at the University of Birmingham. In order to achieve maximum saturation, the samples were initially dried at 80°C for 48 hours, this low temperature was chosen to minimise destruction of the pore-lining clays; electrical measurements on three preserved samples before oven drying and again after drying showed no significant change in the results due to this process. After saturation the cores were left in the same solution to equilibriate for at least two weeks.

[6] The magnitude and the phase angle of the conductivity were measured at a selection of frequencies from 0.0001 to 1000 Hz. Cores were contained in a brine buffered four-electrode cell for measurement of the type used by *Taylor and Barker* [2002] and the brine temperature was maintained at 20 ± 1 °C. Electrodes were non-polarising Ag/AgCl and current density was kept below 0.05 Am^{-2} to avoid non-ohmic effects.

[7] Cation Exchange Capacity (CEC), pore-surface area and Mercury Injection Capillary Pressure (MICP) [*Bloomfield et al.*, 2001; *Lowell and Shields*, 1991] were all measured on small sandstone samples taken immediately adjacent to the core used for the electrical measurements. CEC was measured by saturating exchange sites with NH_4^{+} and measuring released cations with a Philips PU 7450 ICP spectrometer. Pore-surface area was measured using the Micrometrics ASAP 2010 and calculated by a five point Brunauer, Emmett and Teller (BET) method. MICP was measured using the Micrometrics Autopore III. This gives pore-throat size distributions due to the fact that a non-wetting fluid like mercury will not invade a pore-throat unless the fluid exceeds a critical pressure. The size of the pore-throat is inversely proportional to the critical pressure and can be calculated using the Washburn equation [*Bloomfield et al.*, 2001].

3. Sandstone Properties

[8] The sandstones selected were very diverse, from fine grained to coarse grained (0.0625–2 mm diameter) with porosities from 12.4–32.4% and grain densities from 2.19–2.70 gcm^{-3}. The CEC is mostly due to the presence of clays and oxyhydroxides and shows their capacity to exchange cations between the EDL and the pore fluid; this gives an estimate of the quantity of cations in the EDL. The pore-surface area contribution from the grains is very small compared to the large area provided by clays and oxyhydroxides so the surface area gives a measure of the quantity of pore lining materials associated with the EDL. The CEC ranged from 0.79–19.21 meq/100 g and the surface area ranged from 0.42–6.76 m^2g^{-1}. This shows a wide range of clay and oxyhydroxide quantities and a wide range of cation quantities that are associated with the EDL.

4. Results

[9] Complex electrical measurements show that most samples produce a low-frequency phase peak in the range 0.0001–1500 Hz with coarse grained sandstones producing lower frequency phase peaks than fine grained sandstones, a phenomenon that has been observed previously [*Lesmes and Morgan*, 2001]. It can be seen from comparison of thin sections, for example in Figures 2a and 2b, that similar grain sizes can produce very differently sized pore spaces depending on the compaction of the sandstone and any secondary porosity created by the dissolution of grains in situ. Initial examination of the thin sections and comparison with the electrical measurements suggest that similar grain sizes produce lower frequency phase peaks when the grains are less tightly packed and the pore spaces are larger.

[10] The MICP tests were carried out on all 18 samples in order to determine pore-throat size distributions. One of the samples had a low nearly constant phase spectra and was also characterised by low CEC and pore-surface area. Most of the samples showed striking comparisons between the pore-throat size distributions and the phase spectra. Four pore-throat size distributions, which are representative of the different distributions measured (Figure 3a), are shown here for comparison with the phase spectra of the same samples (Figure 3b). The dominant pore-throat size is shown by the steep rise in the quantity of mercury intruded into the sample (Figure 3a). It can be seen that the higher the frequency of the phase peak (Figure 3b) the smaller the dominant pore-throat size (Figure 3a). Also, the more dominant the pore-throat size, the sharper the phase peak. When there is no clear dominant pore-throat size the phase peak is seen to have a broad spread (Figures 3a and 3b). The dominant pore-throat size is shown against the phase peak value for all samples with both a phase peak and a dominant pore-throat size (Figure 4). The level of correlation between the pore-throat size and the quadrature conductivity within the samples measured so far allows dominant pore-throat sizes (D (mm)) from 0.01–0.06 mm to be calculated from the phase peak frequency (f_{peak} (Hz)) to within 0.004 mm standard error (equation (1)), which strongly suggests that it

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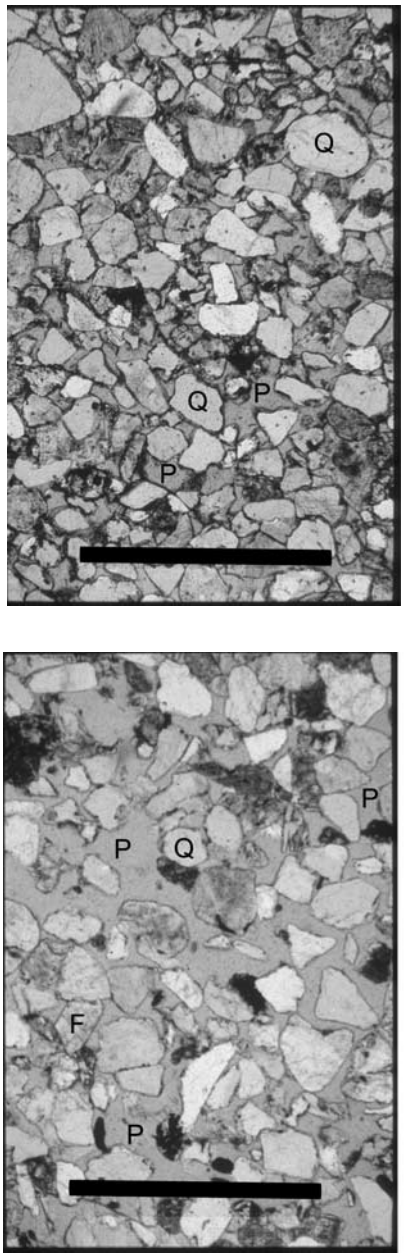


Figure 2. Thin sections of Triassic sandstone samples with 1 mm black scale bar. (a) Sample from Kirkham, Northwest England. An aquifer rock associated with the East Irish Sea basin. (b) Sample from Selby, Northeast England. An aquifer rock associated with the North Sea Basin. Pore space (P), Quartz grain (Q), Feldspar (F).

is the pore-throat size that is actually dictating the relaxation time.

$$D = 0.0162 - 0.0059\ \mathrm{Ln}\left(f_{peak}\right) \quad (1)$$

5. Discussion

[11] We propose a possible model for the low-frequency conduction in sandstone where most conduction takes place within the pores. This conduction is in-phase and carried by

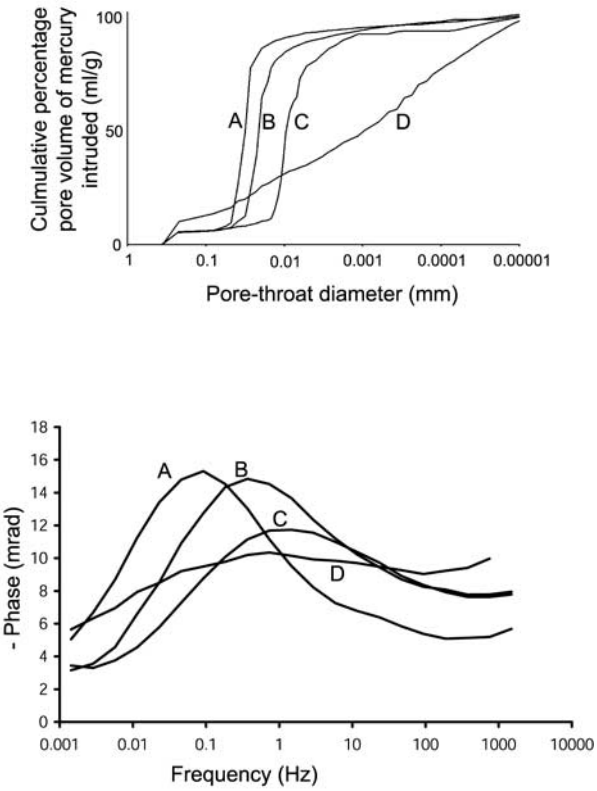


Figure 3. A comparison of pore-throat size with phase spectra of four Triassic sandstone samples A, B, C and D, taken from a 50 m borehole at the University of Birmingham, central England. (a) MICP cumulative pore-throat size distribution. (b) Low-frequency phase spectra from electrical measurements.

ions in the brine. There is also some excess, complex conductivity, along pore walls through the EDL. Within the pore-throats a greater proportion of the conductivity is carried through and adjacent to the EDL so the quadrature contribution is likely to be larger from the pore-throats. Our results suggest that the out-of-phase relaxation mechanisms could be dominated by the pore-throats, with the size of the

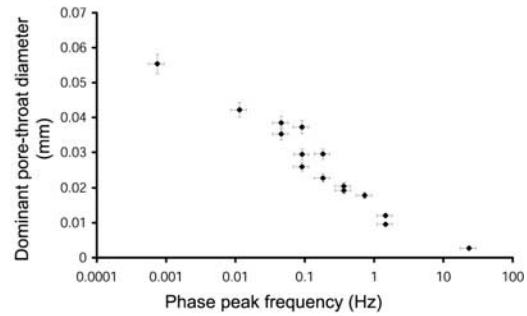


Figure 4. Dominant pore-throat diameter from MICP against the frequency value of the phase peak from electrical measurements. The results of all sandstones measured that had a dominant pore-throat size and a well-defined phase peak.

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pore-throats and smaller connecting pore spaces giving rise to the observed relaxation times. The large range of CEC values and pore-surface areas for the samples suggest that different EDL properties caused by different clay quantities and types have a much lower significance on relaxation time in the Permo-Triassic sandstones of the U.K. than the size of the pores and pore-throats. However, different clay quantities significantly affect the magnitude of the quadrature conductivity.

[12] Saturating with a NaCl brine and increasing the salinity concentration from 0.5 g/l to 4.0 g/l on four of the samples gave a significant decrease in phase angle. This is expected as the in-phase conduction in the brine becomes dominant. However, the phase peak position shifted slightly to a higher frequency. The maximum shift measured for this salinity increase was a doubling of the peak frequency.

[13] Anisotropy, the change of electrical properties with direction of current flow, has not been fully investigated in this study but was not noted to be significant on two of the samples taken perpendicular to each other. Anisotropy may have caused some scatter of the results and has the potential to give important information on the shape of the pore space and should be investigated further.

[14] We believe that previous conductivity models based on grain size distributions [*Lesmes and Morgan*, 2001; *Nettelblad and Niklasson*, 1997; *Garrouch*, 1999] need to be re-examined in the low-frequency range. It is not possible to directly apply the models for relaxation at over 1000 Hz around colloidal particles (<1 μm diameter - clay sized particles) [*Lykema et al.*, 1983] to low-frequency conduction around tightly packed grains (>0.0625 mm diameter - sandstone grain size). Pore-throats and pore geometry are known to contribute significantly to both in-phase and out-of-phase conduction at low frequencies [*Titov et al.*, 2002; *Lima and Niwas*, 2000; *Moss et al.*, 2002] and any electrical model would be incomplete without the inclusion of the pore-throats and pore geometry. We have shown that the pore size is not always linked to the grain size in our samples (Figure 2) and it is known that the pore-throat size is not always dependent on the pore size [*Bloomfield et al.*, 2001]. Therefore this could be the reason grain size distribution models [*Lesmes and Morgan*, 2001] only give an approximation over a limited frequency range to the observed electrical phenomena in sandstone. The ability of low-frequency quadrature conductivity to give pore-throat size distributions is an important step forward in understanding the electrical conduction in sandstone. It has a wide range of practical applications for the hydrocarbon and groundwater industries including permeability estimation and the understanding of hydrocarbon migration within reservoir rocks.

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R. D. Barker and J. B. T. Scott, School of Geography, Earth and Environmental Sciences, University of Birmingham, Edgbaston, Birmingham B15 2TT, UK. (j.b.t.scott@bham.ac.uk)

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